Lecturers: Thomas Quella and Nick Beaton
School of Mathematics and Statistics

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University of Melbourne

# MAST20030: Differential Equations
# Tutorial Solutions: Week 2

**Solution to Question 1.**

(a) To find the general solution, we compute the integrating factor

$$I(x) = \exp\left[\int \frac{1}{t}\mathrm{d}x\right] = \exp[\log t] = t$$

Multiplying the ODE by $I(x)$ gives

$$\begin{aligned} ty' + y &= t\mathrm{e}^t \\ \frac{\mathrm{d}}{\mathrm{d}t}(ty) = t\mathrm{e}^t &= t\mathrm{e}^t - \int \mathrm{e}^t \mathrm{d}t = t\mathrm{e}^t - \mathrm{e}^t + C \qquad \text{(using integration by parts)} \\ y &= \frac{t-1}{t}\mathrm{e}^t + \frac{C}{t} \end{aligned}$$

(b) i. The linear operator is

$$\mathsf{T} = \frac{\mathrm{d}}{\mathrm{d}t} + \frac{1}{t}$$

ii. The complimentary solution, i.e., the solution space of $\mathsf{T}$, is

$$\frac{C}{t}.$$

A particular soltion to the ODE is

$$\frac{t-1}{t}\mathrm{e}^t.$$

Note this is only one option for the particular solution (found by setting $C = 0$). Any other value of $C$ would give a different particular solution, for example $\frac{t-1}{t}\mathrm{e}^t + \frac{2}{t}$ is also a valid particular solution.

iii. The complimentary solution contains one linearly independent function and so the dimension of the solution space of $\mathsf{T}$ is one.

**Solution to Question 2.**

(a) Substituting $\mathrm{e}^{\lambda t}$ into the ODE gives

$$\frac{\lambda^2}{2}\mathrm{e}^{\lambda t} + \sqrt{2}\lambda\mathrm{e}^{\lambda t} + \mathrm{e}^{\lambda t} = 0 \qquad \Rightarrow \qquad \frac{\lambda^2}{2} + \sqrt{2}\lambda + 1 = 0 \qquad \Rightarrow \qquad \frac{1}{2}\left(\lambda + \sqrt{2}\right)^2 = 0$$

so $y = \mathrm{e}^{-\sqrt{2}t}$ is a solution to the ODE.

(b) Substituting $y(t) = u(t)\mathrm{e}^{-\sqrt{2}t}$ into the ODE gives

$$\begin{aligned} \frac{1}{2}\left(u''(t)\mathrm{e}^{-\sqrt{2}t} - 2\sqrt{2}u'(t)\mathrm{e}^{-\sqrt{2}t} + 2u(t)\mathrm{e}^{-\sqrt{2}t}\right) + \sqrt{2}\left(u'(t)\mathrm{e}^{-\sqrt{2}t} - \sqrt{2}u(t)\mathrm{e}^{-\sqrt{2}t}\right) + u(t)\mathrm{e}^{-\sqrt{2}t} &= 0 \\ \frac{1}{2}\left(u''(t) - 2\sqrt{2}u'(t) + 2u(t)\right) + \sqrt{2}\left(u'(t) - \sqrt{2}u(t)\right) + u(t) &= 0 \\ \frac{1}{2}u''(t) &= 0 \end{aligned}$$

and therefore $u(t) = C_1 + C_2 t$. This leads to the general solution $y(t) = C_1\mathrm{e}^{-\sqrt{2}t} + C_2 t\mathrm{e}^{-\sqrt{2}t}$.

(c) Let $y_1(t) = \mathrm{e}^{-\sqrt{2}t}$ and $y_2(t) = t\mathrm{e}^{-\sqrt{2}t}$. The Wronskian is therefore

$$\begin{aligned}\operatorname{wr}\left[\mathrm{e}^{-\sqrt{2}t}, t\mathrm{e}^{-\sqrt{2}t}\right] &= \det\begin{pmatrix} \mathrm{e}^{-\sqrt{2}t} & t\mathrm{e}^{-\sqrt{2}t} \\ -\sqrt{2}\mathrm{e}^{-\sqrt{2}t} & \mathrm{e}^{-\sqrt{2}t} - \sqrt{2}t\mathrm{e}^{-\sqrt{2}t} \end{pmatrix} \\ &= \left[(1-\sqrt{2}t) - (-t\sqrt{2})\right]\mathrm{e}^{-\sqrt{2}t}\mathrm{e}^{-\sqrt{2}t} \\ &= \mathrm{e}^{-2\sqrt{2}t}\end{aligned}$$

and since there exists a value of $t$ such that the Wronskian is non-zero, the two solutions are linearly independent.

**Solution to Question 3.**

Substituting $y = x^n$ gives

$$\begin{aligned} & n(n-1)x^n - n(x+2)x^n + (x+2)x^n = 0 \\ & \quad \Rightarrow\ (1-n)x^{n+1} + (n^2 - 3n + 2)x^n = 0\ \Rightarrow\ n = 1.\end{aligned}$$

Thus, $y = x$ is a solution. To use reduction of order, we assume a solution form of $y(x) = u(x)x$ so that $y' = xu' + u$ and $y'' = xu'' + 2u'$. We therefore have

$$x^2(xu'' + 2u') - x(x+2)(xu' + u) + (x+2)xu = x^3(u'' - u') = 0.$$

If we let $w = u'$, then $w' - w = 0$ and so $w(x) = A\mathrm{e}^x$. This then gives $u(x) = A\mathrm{e}^x + B$ and so the general solution to the ODE is $y(x) = Ax\mathrm{e}^x + Bx$.

**Solution to Question 4.**

Checking that $y = \mathrm{e}^x$ is a solution of the homogeneous equation amounts to checking that $x - (1+x) + 1 = 0$. To apply reduction of order, we assume that $y = u(x)\mathrm{e}^x$ is a solution. Substituting into the inhomogeneous equation gives

$$x(u'' + 2u') - (1+x)u' = x^2\mathrm{e}^x\ \Rightarrow\ xw' + (x-1)w = x^2\mathrm{e}^x \quad (w = u'),$$

because the terms involving $u$ (but not $u'$ or $u''$) vanish. We therefore get the first-order linear ODE

$$w' + \left(1 - \frac{1}{x}\right)w = x\mathrm{e}^x$$

whose integrating factor is $\exp[x - \log x] = \mathrm{e}^x/x$. Thus,

$$\frac{\mathrm{e}^x}{x}w = \int \mathrm{e}^{2x}\,\mathrm{d}x = \frac{1}{2}\mathrm{e}^{2x} + c\ \Rightarrow\ u' = w = x\left(\frac{1}{2}\mathrm{e}^x + c\mathrm{e}^{-x}\right),$$

where $c$ is a constant. Integrating by parts, we now arrive at

$$u = x\left(\frac{1}{2}\mathrm{e}^x - c\,\mathrm{e}^{-x}\right) - \int\left(\frac{1}{2}\mathrm{e}^x - c\,\mathrm{e}^{-x}\right)\mathrm{d}x = \frac{1}{2}(x-1)\mathrm{e}^x - c(x+1)\mathrm{e}^{-x} + c',$$

where $c'$ is another constant. Finally, this gives the general solution to the inhomogeneous equation:

$$y = u\mathrm{e}^x = \frac{1}{2}(x-1)\mathrm{e}^{2x} - c(x+1) + c'\mathrm{e}^x.$$